Canadian Out-of-district Political Donations (2015 - 2024)*

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Federal political representation in Canada is geographical: one Member of Parliament (MP) represents one electoral district. Despite this representational structure, Canadians can legally donate to a host of political entities, regardless of proximity. Using an open-source political financing dataset compiled by the Investigative Journalism Foundation based on records from Elections Canada, we analyze the extent, timing, and flow of out-of-district donations in Canada from 2015 to 2024. We find that 40% of donor-sourced funding to candidates and district associations comes from outside their district, both during and outside election periods, and that this proportion is consistent across party lines. Outside election cycles, ridings in a number of metropolitan districts send substantially more donations out-of-district than they receive. During election periods, ridings in Brampton, Ontario exhibit particularly large rates of out-of-district donation. On average, slightly over half of out-of-district donations to local political entities are from adjacent districts, while only 17% are from out of province. Interprovincial donations to candidates are significantly more likely from donors in B.C. and Ontario to candidates in Alberta, while interprovincial donations to district associations are more evenly distributed across the provinces.

Keywords: Canadian politics; out-of-district donations; surrogate representation; political participation; political geography

1 Introduction

In Canada, federal representation is structurally geographical; citizens are grouped into a few hundred electoral districts based on where they live, each represented by a single Member

*Code and data are available at: https://github.com/bennypk1/geolocating-political-donations.

of Parliament (MP). Despite this, many citizens seek representation beyond their district's boundaries (Mansbridge 2003), driven by a multitude of motivations, including ideological alignment, partisan loyalty, etc. This "surrogate" representation can manifest itself in many ways, but is primarily exercised through financial contributions, i.e. donations, to political entities outside their district. Out-of-district donations are not always representationally motivated, however. A citizen may donate as a favour to a friend, or to "fit in" with their social circles (Gimpel et al. 2006; Garnett et al. 2022). Given that donations source the majority of funds to campaigns and district associations in North America (Gimpel et al. 2008; Garnett 2024; Elections Canada 2026f), documenting donor patterns and motivations is important in understanding how political influence is distributed across the electorate. While Canadian research exists documenting the extent and patterns of donations in general (e.g. Besco and Tolley 2022; Tolley et al. 2022; Garnett 2024), comparatively little attention has been given to out-of-district donation.

Existing research on out-of-district federal contributions in the U.S., which also uses a geographically-based representational structure, has uncovered substantial rates of out-of-district donation. Both congressional and senatorial candidates presently source around 50-70% of their campaign funding from outside their district, a proportion that has been dramatically increasing over the 21st century (Gimpel et al. 2008; Bednar and Gerber 2011; Sievert and Mathiasen 2023). Incumbent representatives also receive around half of their donations from out-of-district, which has been shown to meaningfully influence their roll-call behaviour (Peoples and Gortari 2008; Canes-Wrone and Miller 2022). These donations are thought to be primarily motivated by partisan gain, evidenced by increased donations to districts with close races (Gimpel et al. 2008; Jacobs and Imboywa 2024). Both parties primarily source local funding from a single small group of highly influential and socio-demographically similar districts (Gimpel et al. 2006).

In this paper, we describe trends in contributions across district lines in the Canadian context, using publicly-available data recording all registered contributions to federal entities from 2015 to 2024. District-level data was matched with socio-demographic data sourced from the 2021 Census, allowing us to better understand why certain districts send and receive as many contributions from others as they do. We find that the candidates and district associations of the average district receive around 40% of all donor-sourced funding from out-of-district, both within and outside election periods. It is our hope that these findings inform further research on effective representation in Canada.

2 Literature Review

2.1 Political Donation, Participation, and Representation

In Canada and many other Western democratic nations, federal representation is fundamentally geographical: a Member of Parliament (MP) is a representative of everyone who lives within the boundaries of their riding. By voting during a general election, the citizens of a given electoral district increase the likelihood that their incumbent MP votes, acts, looks, and thinks how they want them to. Provided that (1) the division of people into ridings efficiently clusters like-minded Canadians, and (2) each Canadian participates equally (for example, through their one vote), we should expect a high degree of congruence between what a legislator does, and what their constituents want.

Beyond voting, individual citizens can increase their local representation by volunteering at a local district association, advocating for their preferred party online, or financially supporting a candidate's campaign (Bednar and Gerber 2011; Baker 2016; Garnett 2024). Since these necessitate more resources (e.g. time, effort, or money), indicate greater political interest, and often have larger impacts on electoral outcomes relative to voting, these constitute "higher forms" of political participation (Milbrath 1981).

Among these, donations stand out as a useful unit for understanding the motivations behind political participation and assessing whether higher forms of political participation distort the dyadic relationship between constituents and their MP. Here, we discuss two reasons why. First, donations are spatiotemporally unconstrained, telling us exactly when and where citizens seek representation (Gimpel et al. 2008; Garnett 2024). While a voter can only seek representation within their district and during an election cycle, a donor can legally contribute to various political entities, whenever, and as often as they want, subject to annual and per-election limits (Gimpel et al. 2008; Elections Canada 2026a). Second, donations are the primary means by which individuals extend their representation beyond district borders. Enabled by modern digital communication, individuals are increasingly motivated to finance national parties, or non-local candidates, in an attempt to be represented on ideological, partisan, or identity-based grounds (Gimpel et al. 2008). For example, a Green party partisan may realize their local candidate will lose, and are content funding another Green candidate in a closer race somewhere else. Similarly, a Black candidate may appeal to Black donors across the country seeking identity-based representation. Given that individual donors are the largest source of campaign funding (Carmichael and Howe 2014; Garnett 2024), high rates of out-of-district donation are of particular concern, since they risk shifting the effective representational structure away from one-legislator-one-district and towards one-legislator-one-donor base.

2.2 Political Donations in the Canadian Context: What has been done

Limited accounts of out-of-district donation rates exist in the Canadian context. An analysis of co-ethnic affinity among donors and candidates found that the likelihood that a donor donates out-of-district is higher for ethnic minorities (50-70%) compared to donors of European origin (33%) (Besco and Tolley 2022). However, these proportions only describe candidate-bound donations, and only during the 2015 general election period. Below we present a review of Canadian research on donations in general, to contextualize our subsequent analysis. We note that this research has focused almost exclusively on donations to candidates during writ periods (i.e. election periods).

Much research has been done to determine the socio-demographic characteristics of donors and candidates in Canada. Most notably, analyses of candidate-bound donations during the 2015, 2019, and 2021 general elections found that donors are more likely to originate from older, wealthier, and better-educated districts (Carmichael and Howe 2014; Garnett 2024). The gender gap in giving has received particular attention, likely given the broader context of the established gender gap in political participation McMahon et al. (2023). From 2004 to 2015, overall federal donation amounts have increased, particularly in Western Canada, and among Conservative political entities (McMahon and Wolfe 2015). However, contribution rates to candidates during the following three general elections (in 2015, 2019, and 2021) have since decreased (Garnett 2024), and the COVID-19 pandemic saw overall reduced political participation (McMahon et al. 2023).

Garnett et al. (2022) propose that donors in Canada donate for two reasons: political (i.e. for representational gain) and transactional (i.e. a favour for a friend, or a willingness to be seen). They find that political motivations for giving are more prevalent overall, but especially among older and affluent donors who contributed larger sums. Transactional donations were more likely at the local level, while political donations were more likely at the national level.

Canadian donors tend to donate more to candidates that they share aspects of their identity with. For instance, women donors are more likely to donate to women candidates (Tolley et al. 2022), and there are high rates of co-ethnic donation, particularly among South Asian donors (Besco and Tolley 2022). More generally, South Asian political participation appears to be particularly heightened in Canada, relative to other ethnocultural groups (Garnett 2024). Together with previous findings on donor motivation, these results suggest that out-of-district donors in Canada are predominantly socially motivated, seeking individual-based representation.

Some work has also been done assessing the impacts of large amounts of political donation on representation. Among incumbent legislators, voting behaviour is seemingly independent of the composition of their donor base (Peoples and Gortari 2008). However, Eagles (2004) found that local campaign donations during the 1993 and 1997 elections meaningfully affected the number of votes a candidate receives.

Finally, we describe some unique aspects of Canadian political engagement, and particularly those that distinguish Canada from the U.S., where the large majority of work on out-of-district donations has been done. Most notably, party loyalty in Canada is weak; it is estimated that only around 2% of Canadians are registered members of any political party (Garnett et al. 2022), compared with around 40% in the U.S. (USAFacts 2026). This has been attributed to many things, including the argument that Canada's multi-party system de-emphasizes partisan membership (Carty 2002). This system also invites more ephemeral loyalties, based on specific policies or identities, and not broad-scope ideological affiliation. Many (historically) large parties in Canada were founded for specific reasons, for example the Bloc Québécois, Reform Party, and Green Party, which cater(ed) to a very specific audience. High rates of immigration may have increase emphasis on cultural and not political belonging (Carty 2002; Besco and Tolley 2022).

2.3 American Out-of-District Donations to U.S. House Representatives

In the U.S., Gimpel et al. (2008) examine out-of-district donations and their impacts on legislative representation, looking at individual out-of-district donations during four Congressional elections between 1994 and 2004. They propose the existence of "donor districts"; a small set of older, wealthy, well-educated, and urban districts wherein a large majority of out-of-district donations originate, destined to Republican and Democrat candidates alike (Gimpel et al. 2008). Notably, these socio-economic predictors of out-of-district donor output in the U.S. are the same as predictors of donor output generally in both the U.S. and Canada (Gimpel et al. 2006; Garnett 2024). Over this time period, they also observed an increase in the proportion of non-local donations received by the average candidate, from around 50% in 1994 to nearly 70% in 2004 (Gimpel et al. 2008). More recent studies suggest that this and similar trends have continued past 2004 (Bramlett et al. 2011; Baker 2016; Canes-Wrone and Miller 2022). Driven by the observation that the "closeness" of local races strongly predicted which districts received these non-local donations, Gimpel et al. argue that gaining a partisan advantage (and not seeking ideological or identity-based representation) is the primary motivation behind non-local donations (Gimpel et al. 2008).

Follow-up U.S. research has found that out-of-district campaign contributions affects the voting behaviour of Congressional representatives, thereby distorting the supposedly-dyadic relationship between a representative and their district. A study of the 2006-2010 Congressional elections revealed that the average House representative becomes less responsive to their constituencies' ideological preferences with increased out-of-district public funding (Baker 2016). Canes-Wrone and Miller (2022) further explore the 2006-2016 Congressional elections to show that House voting behaviour is indeed associated with the preferences of their nation donor base, contrasted against research finding that political action committees (U.S. equivalents to EDAs) do not affect roll call behaviour (Ansolabehere et al. 2003). Another study on senatorial elections emphasizes the rivalrous relationship between constituents and out-of-district donors, experimentally showing that constituents who learned about outside donation expected their senators to be less loyal to local interests (Nathan et al. 2025). More generally, longitudinal analyses of out-of-state (and therefore out-of-district donations) during Senate elections echo many House findings (Sievert and Mathiasen 2023; Jacobs and Imboywa 2024; Keena 2024).

To an even greater extent than in Canada, individual donors are the primary source of funding for both U.S. campaigns and U.S. congressional representatives (Gimpel et al. 2008; Baker 2016). This fact, paired with the many known effects of out-of-district donors on campaigns and representatives discussed above paint the picture that representatives "represent" little more than the whims of their national donor base.

3 Data and Methods

3.1 Data Overview

The data used in this paper was originally sourced from the Elections Canada Open Data website (Elections Canada 2025a). Since 2004, the *Canada Elections Act* has required that all contributions (monetary or otherwise) valued at over $20 be recorded by the recipient and shared with the Government (Elections Canada 2026f). The Elections Canada dataset on donations represents all donations made by individuals to registered federal political entities since that date (Elections Canada 2025a). Additionally, donors who contributed over $200 must have their full name and address (including their postal code) recorded (Elections Canada 2026f). For all contributions, Elections Canada also records the recipient's name, district (if applicable), party (if applicable), entity (e.g. national party, candidate, etc.), and the date the contribution was received. The dataset used in this study is a cleaned version of Elections Canada's, made available to us by the Investigative Journalism Foundation. Their cleaning process is described in Appendix B.1.

Every ten years, federal electoral district (FED) boundaries are redrawn to account for population changes that occurred within the decade (Elections Canada 2025b). We restricted our scope to contributions made during the 2013 representational order, effective between Oct 2nd, 2015 and Apr 22nd, 2024 (Elections Canada 2025b). This is the largest such order present in the full dataset, capturing 61.2% of all recorded contributions since 2004, and spanning three general election periods (42nd, 43rd, and 44th). This decision restricted our focus to a single, static set of FEDs, greatly simplifying our cleaning procedures and analysis. To assess whether or not a donation was out-of-district, donor postal codes were matched with their corresponding FED using Statistics Canada's Postal Code Conversion Files[1] (PCCF), made accessible by the University of Toronto (Statistics Canada, Geography Division 2025). For 22.5% of the total contribution sum[2] during our study period, we were unable to identify the donor's FED. The overwhelming majority (98%) of these unlocated contributions were aggregated sums of anonymized donations under $200. Since donor location is integral to the study of out-of-district donation, these entries were discarded from further analysis[3]. The FEDs of localized recipients (e.g. candidates and district associations) were always present in the dataset.

[1]While nomination contestants are also local political entities as well, they only make up 0.7% of entries and 2.5% of the total amount contributed to local entities. As such, they were not considered to simplify analysis and discussion.

[2]Although we observed that the out-of-district contribution rates varied greatly depending on whether or not the donation was received during an election cycle, we did not split each district into contributions sent during vs. outside election cycles. We did not consider further splits in time since our summaries also showed that metrics were relatively constant within either period.

[3]beta regression is also commonly used to handle outcomes bounded between 0 and 1, and in some cases provides a better fit due to the Beta distribution's handling of heteroskedasticity. Since we observed no heteroskedasticity in our fitted model's residuals, and observed logit-transformed proportions were normally distributed (verified by QQ plot), we thought it fit to use linear regression, for the sake of model simplicity.

Leveraging the availability of donor postal codes, donations were also situated within census dissemination areas (DA), allowing us to attribute socio-demographic characteristics to donation sources at a fine spatial grain. For each dissemination area, we identified its population, population density, median household income, mean age, proportion of the population with a post-secondary education, and proportions of various visible minority ethnicities. A full list of the ethnicities we tracked is available in Appendix C.2. These variables were chosen since they are significantly associated with the propensity to donate during an election (Gimpel et al. 2006, 2008; Garnett 2024). Further, ethnicities were disaggregated since previous Canadian research has found that some minority groups such as South Asians donate at much higher rates than others (Besco and Tolley 2022). We again used the PCCF to match donor postal codes to the DAs of the 2021 Census, and to group DAs into FEDs[4] to deduce the same socio-demographic information at the district level.

Table 1 provides an overview of the primary variables used throughout this study, and an example of each. Census-based variables, as well as geographic attributes such as distance between a donor and the centroid of their recipient's district were easily fetched when needed by relating the `donor_district`, `donor_dissemination_area`, or `recipient_district` variables in Table 1 with Table 2 or Table 3. A detailed walkthrough of the full data pipeline, from the files available for download on Canadian government websites to the core datasets used for analysis can be found in Appendix B and Appendix C.

Most (84.6%) contributions included in our study were addressed to national-level political entities; either party leadership contestants, or parties themselves. Since these entities are not tied to local FEDs, contributions addressed to them cannot be declared as in or out of district. Given their prevalence, however, general trends in these donations are briefly discussed in Section 3.2.1, emphasizing how they mirror or differ from donations to local entities. All data cleaning, visualization, and analysis was conducted using the R statistical software (R Core Team 2025).

[4]While nomination contestants are also local political entities as well, they only make up 0.7% of entries and 2.5% of the total amount contributed to local entities. As such, they were not considered to simplify analysis and discussion.

Table 1: Primary variables used in our study, with examples and descriptions. Although not explicitly shown, additional information used such as census variables or geographic attributes of districts are easily retrived by relating a donor's dissemination area or district to custom-made lookup tables.

Variable	**Example**	**Description**
donor_name	Alexandra Lulka	Donor full name.
donor_dissemination_area	35202387	Dissemination area (UID) associated with donor postal code.
donor_district	Eglinton–Lawrence	Electoral district associated with donor postal code.
donation_date	2015-10-18	Date donation was received.
electoral_event	General Election	Electoral event (if any) occurring during donation.
total_amount	563.3	Summed monetary and non-monetary value of the donation, in CAD.
recipient_name	Joe Oliver	Recipient full name.
recipient_district	Eglinton–Lawrence	District of the recipient (if applicable).
political_entity	Candidates	Entity type of the recipient.
political_party	Conservative Party of Canada	Recipient party affiliation (if any).
is_out_of_district	FALSE	Whether donor and recipient districts are different.

Table 2: Custom spatial lookup table, modified from Statistics Canada's Postal Code Conversion File. This table unambiguously links each Canadian postal code to a 2021 census dissemination area, and each census dissemination area to a federal electoral district from the 2013 representational order. Only a sample of four entries are shown.

Postal Code	**Dissemination Area**	**Federal Electoral District**
M4P1E4	35202387	Eglinton–Lawrence
K1A0A6	35061205	Ottawa Centre
V6B1A1	59153201	Vancouver Centre
H3A0G4	24661048	Outremont

Table 3: Custom census lookup table. This table provides desired socio-demographic information for every dissemination area in the 2021 Census. Only a sample of four entries are shown. Only population, density, and median household income are shown.

Dissemination Area (UID)	Population	Density (per sqkm)	Median Household Income (CAD)
35202387	612	4381.2	$98,500
35061205	548	3920.7	$87,200
59153201	701	6210.5	$76,300
24661048	489	5104.9	$92,100

3.2 Preliminary Data Summaries

3.2.1 Situating Contributions to Local Entities Among All Contributions

We start by providing a set of data summaries intended to situate out-of-district donations among the broader set of federal donations.

Figure 1 displays the breakdown of the total amount contributed during our study period, by type of recipient (i.e. political entity) and whether the contribution was received during an election cycle. Donations received during election cycles represent a minority of all donations (14%), and local and national-level entities receive similar amounts of donor-based funding. Local funding is shared evenly between candidates and registered (i.e. district) associations. Outside of election cycles, national entities (primarily parties but also a non-negligible sum to leadership contestants) receive substantially more donations than local ones. During electoral downtime, notable amounts of local funding are raised only by district associations, meaning that donations to electoral candidates are effectively contained within campaign periods. Both in and out of election cycles, nomination contestants received relatively little amounts of donations, and thus are not shown. Funding distributions are similar when counts were displayed instead of amounts, and when specific events are examined (e.g. 42nd general election, or the period between the 42nd and 43rd general elections).

Table 4 displays the breakdown of total contributed amount by general election period. As expected from Figure 1, the bulk of donations (86%) are received outside general election periods. Among election periods, the 42nd general election received roughly twice as many donations as the others, likely due to its abnormally long writ period. Over all periods, donations to local entities are out-of-district 40-46% of the time.

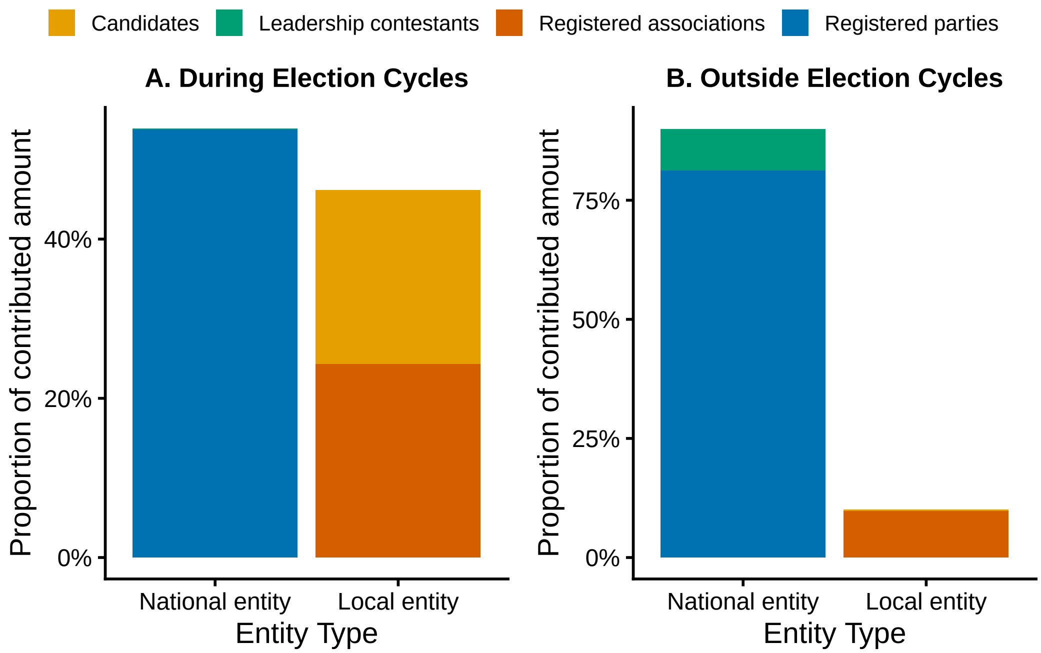

Figure 1: Distribution of contributions by recipient entity (Oct 2015 - Apr 2024), shown during (a) and outside (b) election cycles. Contributions to Nomination contestants were negligibly small, and removed to simplify the visualization.

Table 4: Breakdown of contributions by election period. Proportion of contributed amount sent to out-of-district local entities (OOD %) is also shown for each period.

General Election Period	Amount	% of Total	OOD% (only local entities)
42nd general election	44949473	7%	44.9%
43rd general election	16481907	3%	40.9%
44th general election	25993529	4%	45.6%
None	539255700	86%	46.0%

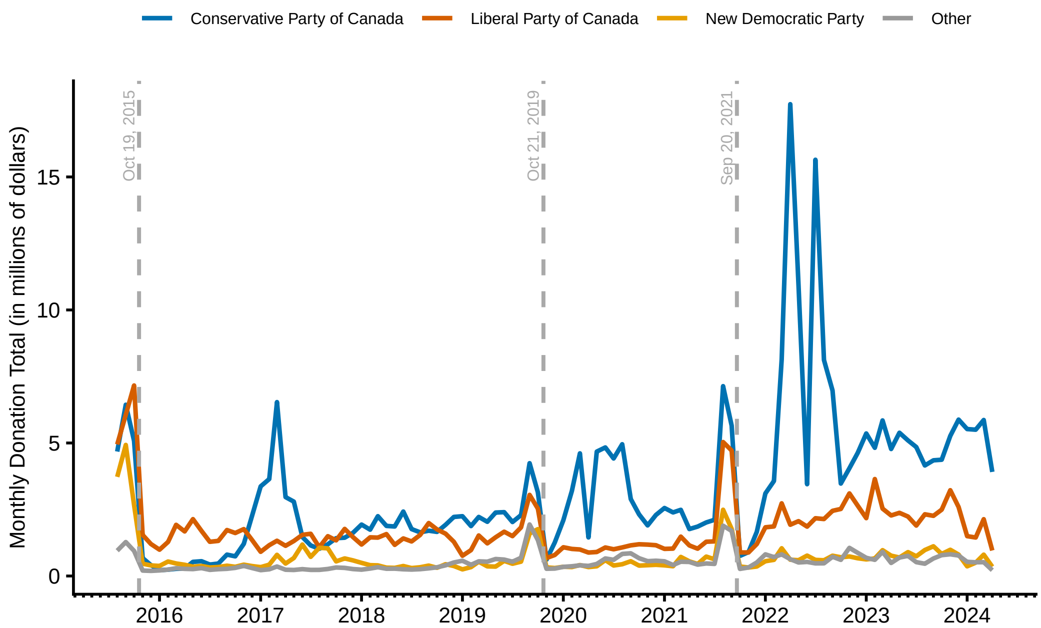

Figure 2: Time series of federal political donations from Oct 2015 - Apr 2024. Trends are separated by the political party of the recipient. Dashed vertical lines indicate general election dates for the 42nd, 43rd, and 44th general elections respectively.

Figure 2 displays the timeline of donations during our study period, separated by the party of the recipient. We briefly note that, for this visualization, donations dated in December of each year were redistributed across the other months of that year to address a clerical error in the original dataset (see Chapter E for more details). The six main peaks in the graph clearly correspond to general election periods (peaks followed immediately by grey dashed lines) and Conservative party leadership campaigns (peaks in the blue trend line). Interestingly, elections and leadership contests result in relatively similar amounts of donations, with the notable exception of the 2022 Conservative leadership race, won by current Opposition Leader Pierre Poilievre. Overall, donations seem to have increased during the study period, following the increasing trend observed from 2004 to 2015 by McMahon and Wolfe (2015), and particularly starting in 2022, driven primarily by increased donations to the Conservative and Liberal parties.

Since the trends in Figure 2 are dominated by national donations, we also display the timeline of contributions to candidates and district associations in Figure 3. As expected from the trends in Figure 2, donations to EDAs and candidates peak during general election periods. Notably, the longitudinal distribution of in and out-of-district donations is seemingly identical, both

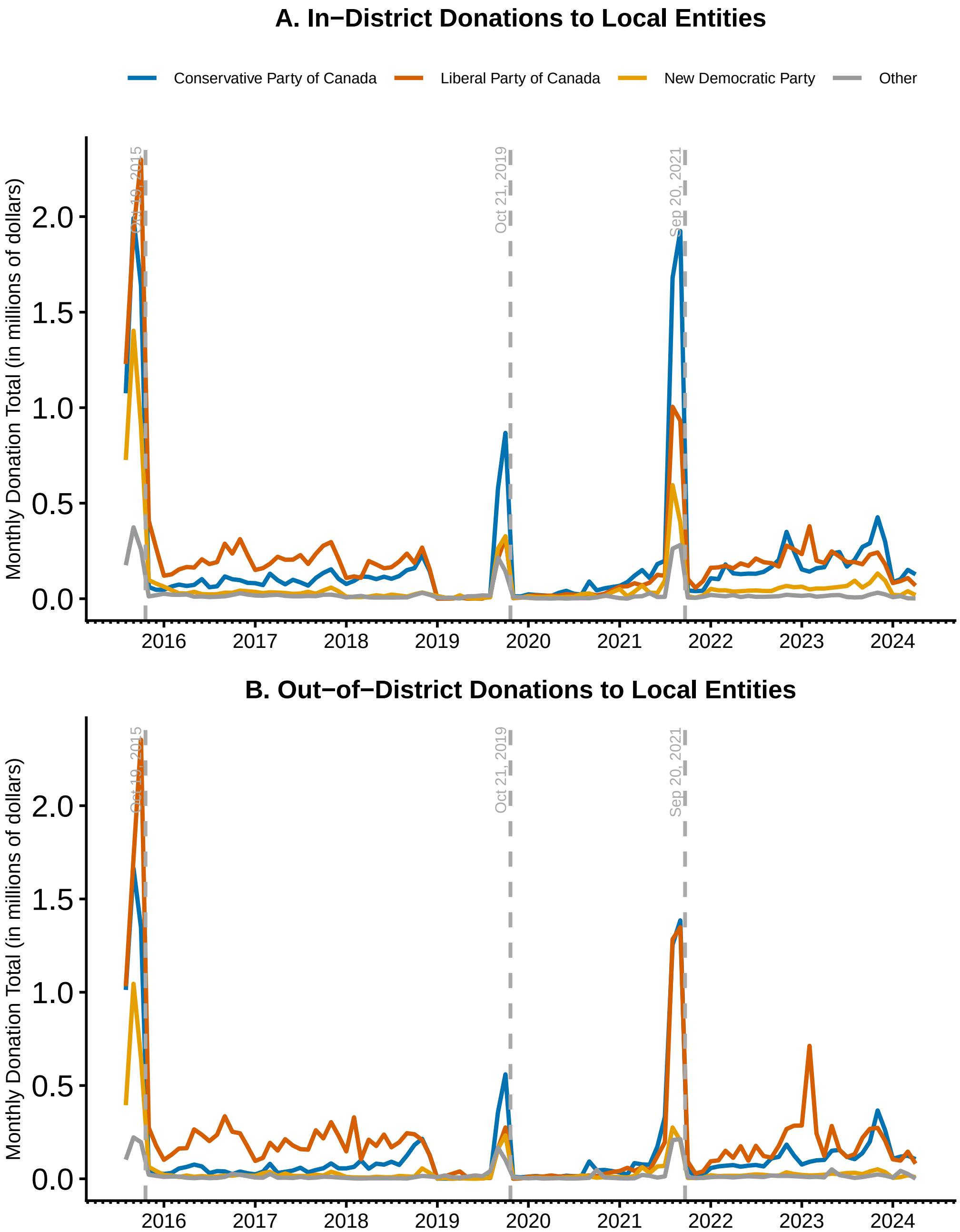

Figure 3: Time series of (A) in-district and (B) out-of-district contributions to local entities (combined candidates and district associations), from 2015 to 2024. Separate trend lines indicate distinct political parties. Dashed vertical lines denote the dates of the 42nd, 43rd, and 44th general elections respectively.

Table 5: Rates of funding sourced from out-of-district (OOD) donors among local political entities.

Entity	Amount	% of Total	OOD %
Candidates (in-cycle)	19088309	20.3%	42.1%
Candidates (out-of-cycle)	1724938	1.8%	51.0%
EDAs (in-cycle)	21209606	22.6%	46.7%
EDAs (out-of-cycle)	51996459	55.3%	45.4%

during and outside general elections, and across the main political parties. We also observe reduced donation amounts in both figures from the start of 2019 to mid-2021, which may be partially explained by the COVID-19 pandemic, and agrees with the findings of McMahon et al. (2023).

Table 5 show some key summary statistics for local political entities. Both in and out of election cycles, candidates and EDAs receive 42-51% of their donor-sourced funding from outside their district. Given how similar in-district and out-of-district contribution trends are over time (Figure 3), the annual means presented in Table 5 are likely representative of the out-of-district proportions at any specific point in time. While the out-of-district funding proportion in the U.S. has dramatically increased over time, we observe no evidence of this among Canadian donors.

Through these preliminary summaries, we discover that consistent national-level donations dominate trends in contributions over time; party leadership contests create swells of contribution, comparable in size to those of general elections, and the top 3 parties claim both a majority of donors, and of total contributed amount. Despite this, local EDAs (and candidates during election periods) form a consistently non-negligible fraction of donation recipients, even outside election periods. Further, nearly half of the funds raised by EDAs and/or candidates originate from donors located outside the boundaries of their respective districts, both during and outside election cycles (Table 5). Together these findings help to situate further analysis of out-of-district federal donations within the broader context of federal donations in general.

3.2.2 General Trends in Out-of-District Contributions

Here we document general trends in out-of-district donations. While the above section focused on individual contributions, in this section we present statistics and visualizations at the

Table 6: Proportion of donations sent vs. received by the average district from out-of-district, separated by the type of receiving political entity.

Statistic	Value
Mean Proportion of Candidate-Bound Contributions Received from Out-of-District	39.8%
Mean Proportion of Candidate-Bound Contributions Sent Out-of-District	43.0%
Mean Proportion of EDA-Bound Contributions Received from Out-of-District	41.9%
Mean Proportion of EDA-Bound Contributions Sent Out-of-District	42.2%

Table 7: Proportion of donations received by the average district from out-of-district (OOD), separated by the type (candidate vs. entity) and party of the receiving political entity.

Party	Entity	% Donations Received from OOD
Conservative Party of Canada	Candidate	39.5%
Conservative Party of Canada	EDA	39.6%
Green Party of Canada	Candidate	36.3%
Green Party of Canada	EDA	30.6%
Liberal Party of Canada	Candidate	38.5%
Liberal Party of Canada	EDA	40.1%
New Democratic Party	Candidate	38.5%
New Democratic Party	EDA	35.3%

district-level. We only consider donations to local political entities[5] (i.e. candidates and district associations), since these are the only recipient types where contributions can be assessed as out-of-district. These represented 9.7% of entries and 15% of the total contributed amount in the dataset. Of all possible pairs of sending and receiving districts, only 11.7% (n = 13,420) had even a single local entity-bound contribution recorded in the dataset.

Table 6 and Table 7 display proportions of out-of-district funding sent and received by the average ridings. On average, a district sends and receives around 40% of all local contributions to and from other districts. Table 6 breaks down this proportion by entity (candidate vs. riding) and Table 7 stratifies receiving donations by political party. In all cases, the proportion of out-of-district funding received by the average district ranges between 30.6% (the proportion received by Green party EDAs, for the average district) and 43% (the proportion sent to

[5]While nomination contestants are also local political entities as well, they only make up 0.7% of entries and 2.5% of the total amount contributed to local entities. As such, they were not considered to simplify analysis and discussion.

candidates, for the average district).

Figure 4 compares how districts are distributed in terms of the amount of contributions they receive vs. the amount of contributions they send to and from candidates and district associations. Since the total amount sent and received must be equal, the average district sends and receives the same amount of contributions, hence the shared mean in the top and bottom subplots. The distribution of senders to district associations is particularly skewed, indicating that there exists a smaller subset of districts that send much larger sums to these local entities than the rest. While the receiving distribution is also skewed, it is visibly less so, indicating that a larger number of districts receive near the average proportion of contributions from outside their district to their district associations. Interestingly, this is not observed among out-of-district donations to candidates; distributions of senders and receivers are expectedly skewed, however the extent of the skew is very similar for senders and receivers.

Figure 5 displays the relationship between the amount sent and received from other districts, for each of the FEDs in our study. Subplots separate the data into donations to EDAs vs. donations to candidates. For EDAs, the top senders and receivers fall cleanly into two groups: those that send substantially more than they receive (e.g. University–Rosedale), and those that receive substantially more than they send (e.g. Scarborough Centre). For candidates, 4 of the 5 top senders/receivers are the districts that compose the Brampton region in the Greater Toronto Area. We observe a similar separation between top senders and receivers.

Figure 6 displays the relationship between the amount donated in vs. out-of-district, for each of the FEDs in our study. Subplots separate the data into donations to EDAs vs. donations to candidates. Across both entities, the top out-of-district contributors donate more to other districts than the top in-district contributors donate to themselves. For donations to candidates, most of the top contributing FEDs donate mostly within their own districts, with the notable exception of 6 FEDs, the top 3 of which are in Brampton. Together, Figure 5 and Figure 6 demonstrate how varied FEDs are in the amount of funding they allocate to other districts (especially to candidates), and the amount sent vs. received.

Table 8 and Table 9 further breakdown where out-of-district donations themselves are sourced and received. We observe that FEDs on average send and receive roughly half of their out-of-district donations to/from adjacent districts. This tells us that donations are only partly spatially constrained. Donations are substantially less likely to finance candidates or district associations outside provincial boundaries, with only around 15% of funds being sent or received cross-provincially. Comparing the two tables, inter-provincial and non-adjacent donations

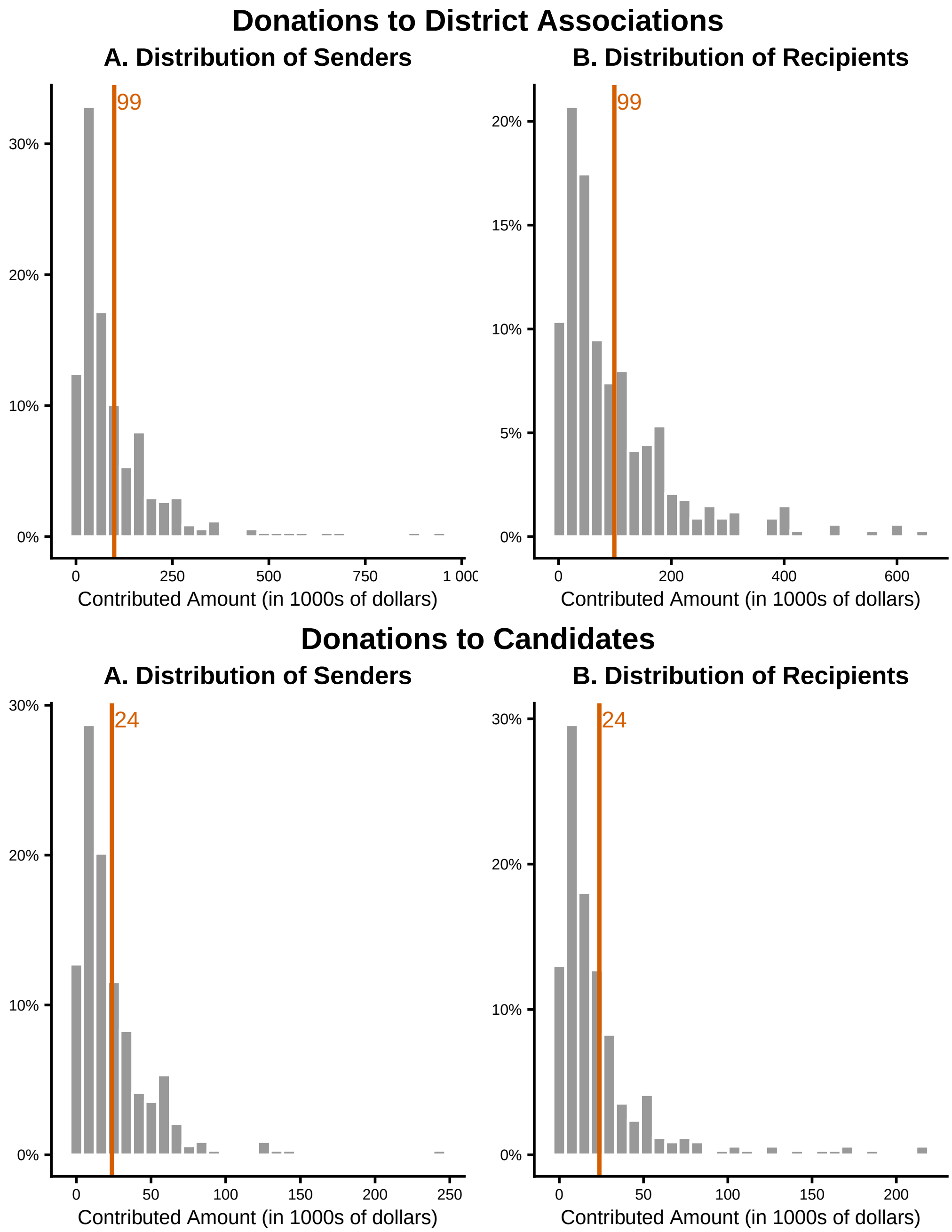

Figure 4: Distribution of districts by total amount (A) sent out-of-district and (B) received by donors in other districts, from 2015 to 2024. The top two subplots display these distributions for donations to district associations, while the bottom two display the same distributions but for donations to electoral candidates. Orange lines indicate the means of their respective distributions.

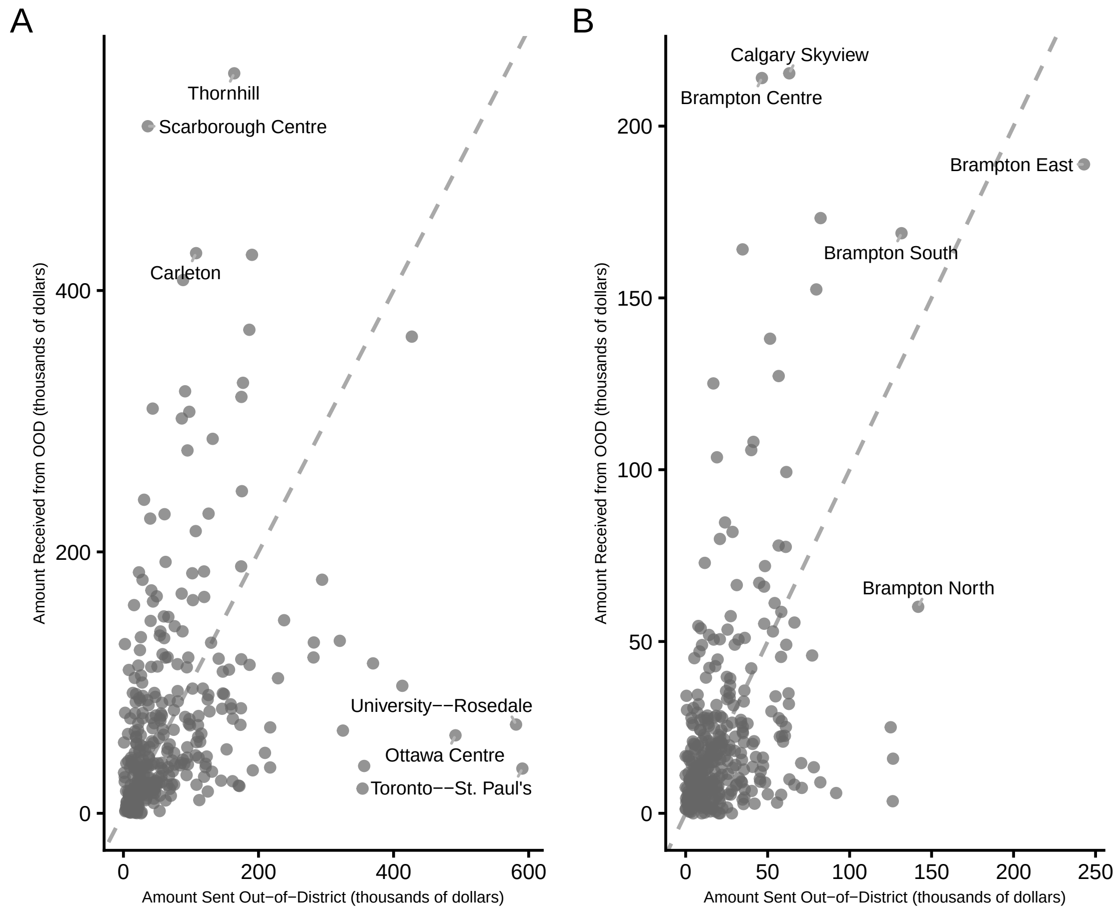

Figure 5: Relationship between the contribution amount a district sends and receives to/from other districts, for each district during the 2013 representational order (effective from 2015 - 2024). First subplot (A) shows this for donations to EDAs, and second (B) shows this for donations to candidates. Grey dotted line is 1:1, meaning that points on this line receive and send the same amount of donations. Top three sender and receiver FEDs are labelled with their names.

Table 8: Of all out-of-district (OOD) contributions to district associations (EDAs), the mean proportion that also leave the province or land in a non-adjacent district, measured per district from both the sending and receiving side.

Statistic	Direction	Value
Mean Proportion of OODs from Out-of-Province	Sent	14.7%
Mean Proportion of OODs from Out-of-Province	Received	16.5%
Mean Proportion of OODs from Non-Adjascent Districts	Sent	56.9%
Mean Proportion of OODs from Non-Adjascent Districts	Received	58.7%

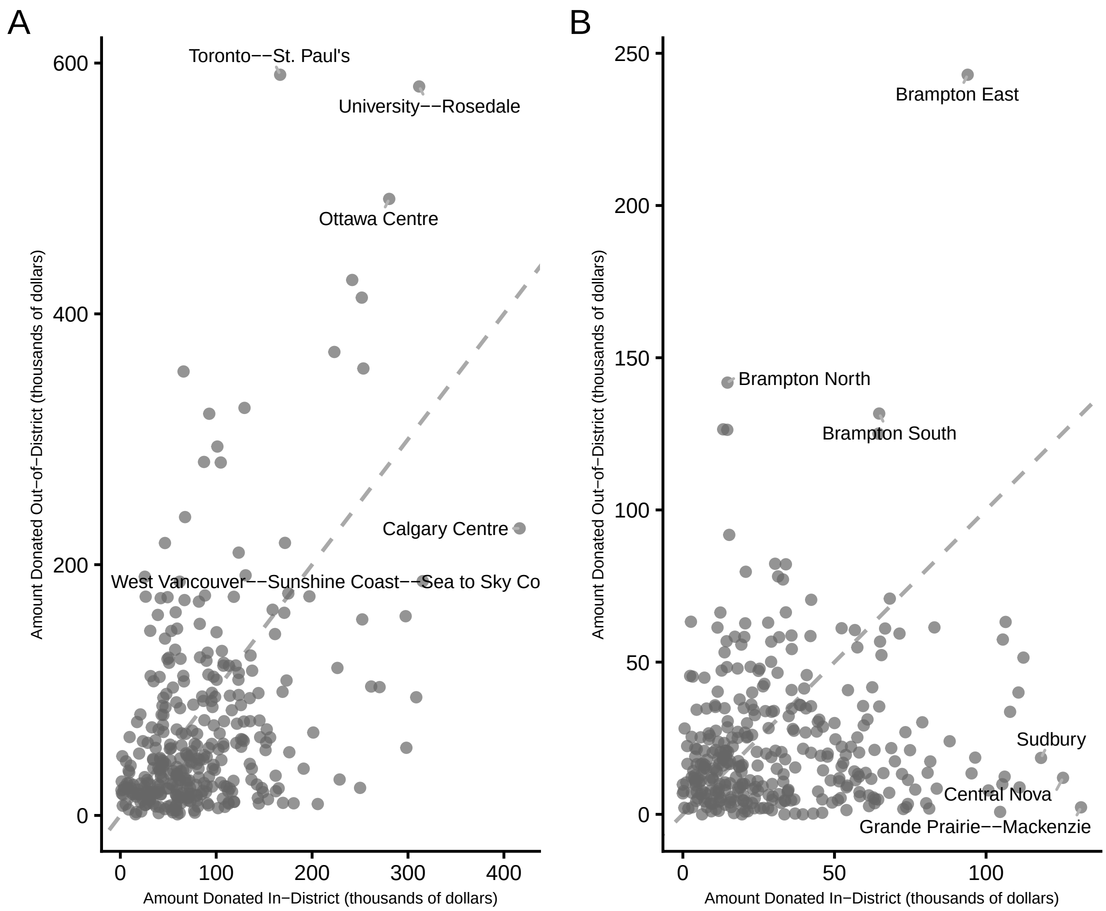

Figure 6: Relationship between the amount donated in v.s. out of district, for each district during the 2013 representational order (effective from 2015 - 2024). First subplot (A) shows this for donations to EDAs, and second (B) shows this for donations to candidates. Grey dotted line is 1:1, meaning that points on this line send the same amount of donations in vs. out-of-district. Top 3 in-district and out-of-district donors are labelled with their names.

Table 9: Of all out-of-district (OOD) contributions to electoral candidates, the mean proportion that also leave the province or land in a non-adjacent district, measured per district from both the sending and receiving side.

Statistic	Direction	Value
Mean Proportion of OODs from Out-of-Province	Sent	12.7%
Mean Proportion of OODs from Out-of-Province	Received	10.4%
Mean Proportion of OODs from Non-Adjacent Districts	Sent	53.0%
Mean Proportion of OODs from Non-Adjacent Districts	Received	53.0%

are more common for contributions addressed to EDAs compared with those addressed to candidates during election periods (Table 8).

Although Table 8 and Table 9 tell us that out-of-district donations are largely contained within the same province, we display these inter-provincial flow patterns in Figure 7. For district associations (where out-of-province contributions make up a larger proportion of out-of-district contributions), Ontario, B.C., and Quebec are the major sources of flux. Generally, total flux seem to closely correspond to the population size of the province. B.C., Ontario, and Alberta give more than they receive, while the rest receive more than they give. The largest flux is the contributed amount given to Québécois EDAs by Ontario donors. Strikingly, donations to candidates are dominated by completely different links. Ontario and B.C. remain the primary net givers, but their contributions target candidates in the prairies. Given that most donors donate to party-affiliated entities (see Figure 1 above), this may be explained by partisan differences between the three provinces; liberals in Ontario and B.C. may be donating to liberals in Alberta. Interestingly, these flows are not reciprocated to nearly the same extent, even though EDA fluxes are.

3.3 Modelling Out-of-District Propensity

3.3.1 Model Specification

Inspired by the large variation in out-of-district giving observed in Figure 6, we fit a linear regression model to characterize what features of a district are associated with higher rates of out-of-district donation. Our outcome variable is the share of all contributions that were sent out-of-district during our entire study period[6], for each district, which was logit-transformed to remove its boundedness[7]. Since there are 338 districts in the 2013 representational order, our model was fit on that number of observations.

We considered the following set of covariates: population density, median household income, mean age, the proportion of the population with a post-secondary degree, total contribution

[6]Although we observed that the out-of-district contribution rates varied greatly depending on whether or not the donation was received during an election cycle, we did not split each district into contributions sent during vs. outside election cycles. We did not consider further splits in time since our summaries also showed that metrics were relatively constant within either period.

[7]beta regression is also commonly used to handle outcomes bounded between 0 and 1, and in some cases provides a better fit due to the Beta distribution's handling of heteroskedasticity. Since we observed no heteroskedasticity in our fitted model's residuals, and observed logit-transformed proportions were normally distributed (verified by QQ plot), we thought it fit to use linear regression, for the sake of model simplicity.

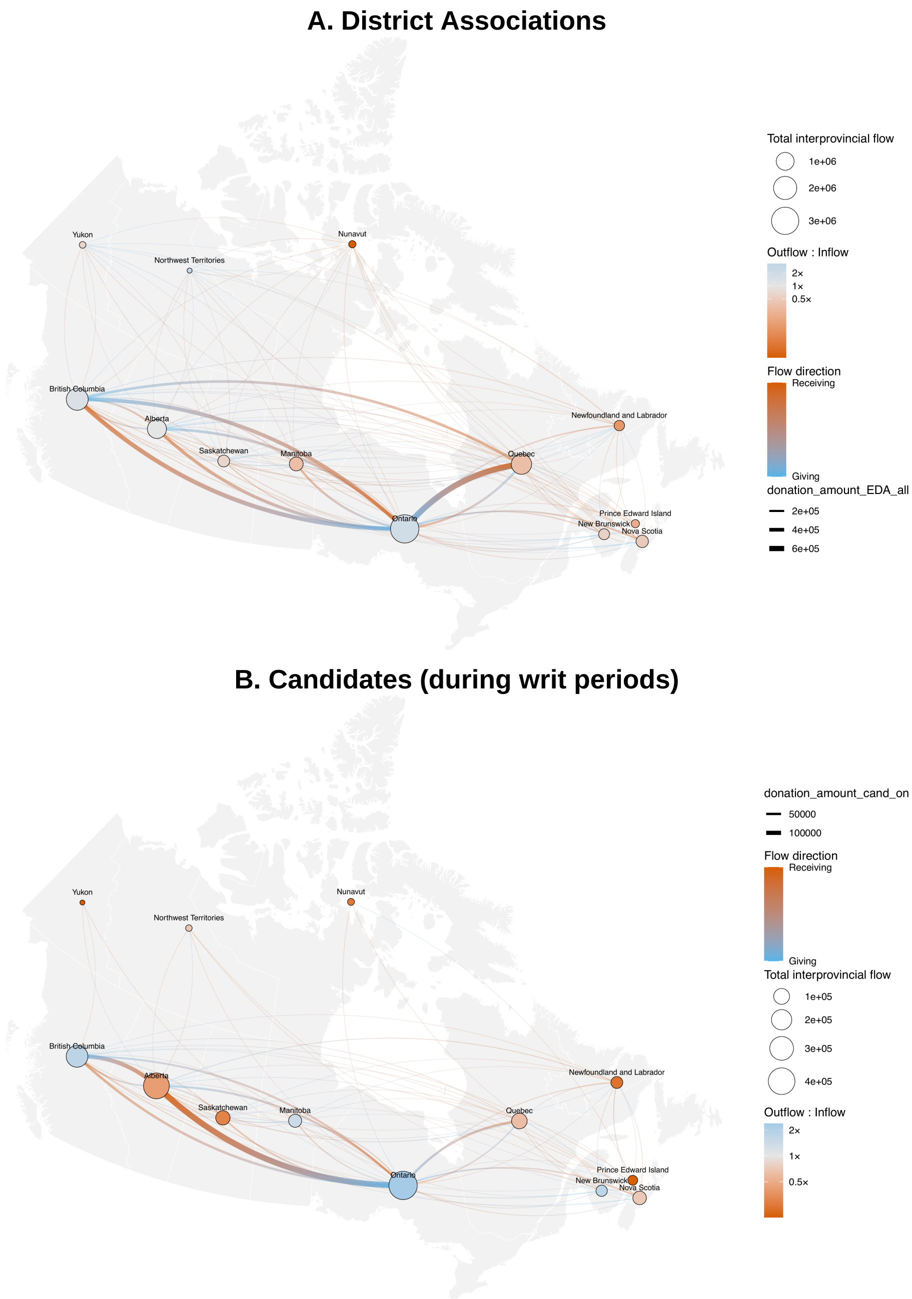

Figure 7: Network of out-of-district contributions to (A) local district associations from 2015 to 2024 and (B) electoral candidates during the 2015, 2019, and 2021 writ periods. Red nodes indicate provinces that received more contributions than they gave (vice-versa for blue nodes). Node size indicates the combined total amount of contributions sent and received by a province. Donations flows are colored on a gradient from blue (the sender) to red (the recipient), sized according to the amount flowing from one province to another.

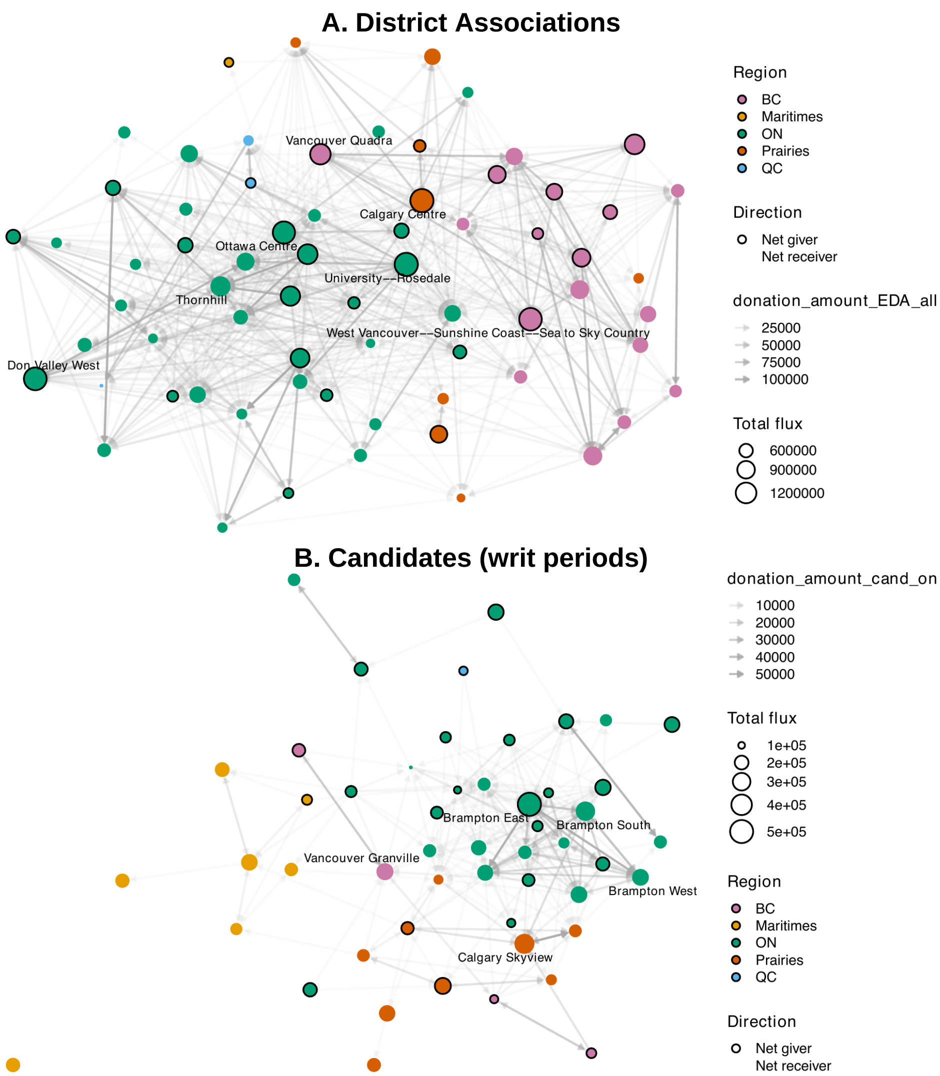

Figure 8: Network of out-of-district contributions to (A) local district associations from 2015 to 2024 and (B) electoral candidates during the 2015, 2019, and 2021 writ periods. Figure includes only the 20% of districts with the largest total flux (combined amount given and received). Nodes with a black outline indicate net-donor districts, while borderless nodes indicate net-receiver districts.

amount, and the proportion of the population that is a visible minority. Population density was considered for two reasons. First, we suspected that districts clustered together in space (for example, the districts of downtown Toronto) would have higher rates of out vs. in-district donation than the average district, simply due to spatial proximity. Since districts are constructed to have relatively similar population sizes (Elections Canada 2025b), population density serves as a suitable proxy for how spatially clustered a district is. Second, densely populated districts are much more likely to be in affluent areas, which have increased out-of-district giving rates in the U.S. (Gimpel et al. 2008). Total contribution amount was also included for this latter reason. Median household income, mean age, the proportion of the population with a post-secondary degree were also included for this reason, but also since Canadian research has shown that these are significant predictors of donation rates in general (Garnett 2024). Finally, visible minority proportion was included since, U.S. scholars have found that this decreases out-of-district donor propensity (Gimpel et al. 2006, 2008), while Canadian scholars have found that it increases it (Besco and Tolley 2022). Due to moderate levels of multicollinearity between these socio-demographic variables, we were careful about our choice of parameters to definitively include, to avoid diluting individual variables' significance. Also, certain variables were transformed to improve linearity, while still preserving interpretability.

Our final fitted model is presented in Section 3.3.2, and a full discussion of variable selection and assumption verification is presented in **Appendix X**.

3.3.2 Model Results

Our chosen model structure is presented here (with coefficients shown in Table 10):

$$
\begin{aligned}
\operatorname{logit}(p_i) = \beta_0 + \beta_1\sqrt{\text{Density}_i} + \beta_2\text{Age}_i + \beta_3\log(\text{Income}_i) \\
+ \beta_4\log(\text{Amount}_i) + \beta_5\operatorname{logit}(\text{VisMin}_i) + \beta_6\operatorname{logit}(\text{PostSec}_i) + \epsilon_i
\end{aligned}
$$

where:

- p_i is the share of the total contributed amount (to candidates and district associations) that was sent out-of-district by district i.
- Density_i is the population density of district i (persons per km^2).
- Age_i is the average age of residents in district i.
- Income_i is the median household income in district i.

Table 10: Coefficients, standard errors, t-statistics, and p-values for the fitted out-of-district donation propensity model. The response is the logit-transformed share of the total contributed amount sent out-of-district.

Term	Coefficient	Std. Error	t-statistic	p-value
Intercept	-10.027	3.457	-2.90	0.00397
Population density (square root)	0.011	0.003	4.07	5.78e-05
Average age	0.060	0.016	3.88	0.000128
Median household income (log)	1.297	0.280	4.64	5.1e-06
Total amount donated (log)	-0.634	0.063	-9.98	<2e-16
Visible minority proportion (logit)	0.406	0.044	9.13	<2e-16
Post-secondary proportion (logit)	0.903	0.143	6.30	9.23e-10

- Amount_i is the total amount contributed by donors in district i.
- VisMin_i is the proportion of residents in district i who identify as a visible minority.
- PostSec_i is the proportion of residents in district i who hold a post-secondary qualification.
- ϵ_i is the error term for district i.

Residual analyses showed that our model clearly satisfies the four core assumptions of linear regression (see Figure 9). VIF-analysis confirmed that multicollinearity among predictors still exists, but no scores were over 5, so this was not considered to be a major issue. All covariates considered had significant effects on the logit-transformed of out-of-district donation propensity. All effects were expectedly positive, apart from the total amount donated.

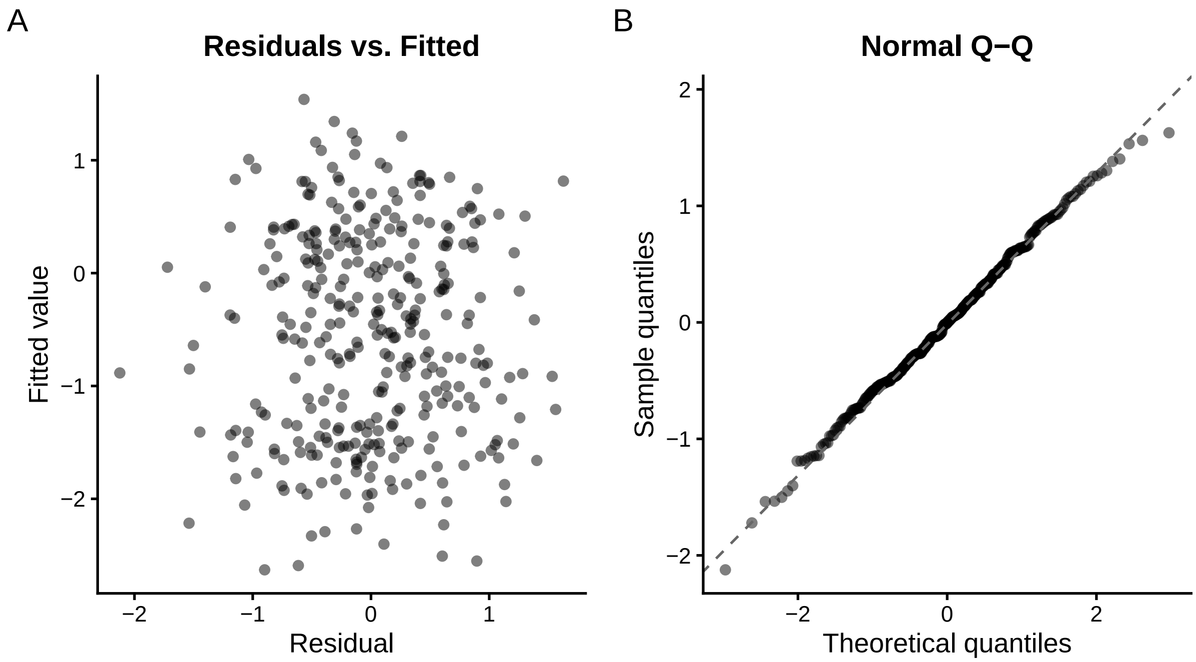

Figure 9: Diagnostic plots for the fitted out-of-district donation propensity model. (a) plots residuals against fitted values. (b) shows a Normal Q-Q plot of the model residuals, with the dashed line indicating the theoretical normal quantiles.

4 Discussion

Using political financing data compiled by Elections Canada, we examined the extent and spatial flow patterns of out-of-district donations to federal candidates and electoral district associations (EDAs) between 2015 and 2024. Here we present a synthesis of our main findings, and discuss notable study limitations.

Regardless of party affiliation, or whether an election is officially underway, the average electoral candidate and district association (EDA) both receive around 40% of their donations from Canadians residing in other districts. However, donations to EDAs and candidates exhibit qualitatively distinct flow patterns at the district scale. Out-of-district donations to EDAs are led by a small group of districts (e.g. University Rosedale, Trinity St. Paul, Calgary Centre), both by proportion and gross amount donated. Districts contributing the largest total amounts also tend to direct a substantial share of their EDA donations to organizations outside their own district. The relative affluence of these source districts is consistent with the primary characteristic of "donor districts" identified in previous North American studies of political contributions. (Gimpel et al. 2006, 2008; Garnett 2024; Jacobs and Imboywa 2024). By contrast, donations to candidates are driven by a different cast of districts; the four Brampton constituencies in Ontario. While candidate-bound donations from the average district are much more localized than donations to EDAs, the donors of North, South, East, and Central Brampton dominate the out-of-district candidate donations network, sending comparatively massive amounts of funding to candidates in adjacent Brampton districts. These patterns are consistent with previous research documenting the prevalence of co-ethnic political donations (Besco and Tolley 2022; Tolley et al. 2022), and the striking overrepresentation of South Asian donors in the Canadian federal contributions scene (Garnett et al. 2022).

Our analysis of interprovincial donations reveals yet another dimension of Canadian donor behaviour. In contrast to the increasing nationalization of out-of-district donations in the U.S. (Jacobs and Imboywa 2024), the majority (~80%) of out-of-district donations in Canada are contained within provincial boundaries, regardless of the recipient's party, and especially during election cycles. However, those that breach these boundaries exhibit drastically different patterns depending on recipient type. Donations to EDAs in separate provinces follow similar patterns as their superset; sources, sinks, and flow magnitudes are largely explained by differences in wealth and population size. However, while out-of-district donations to electoral candidates are primarily motivated by ethnic affinity, the small subset of these that are out-of-province is better explained by partisan motivations. Out-of-province donations to candidates are dominated by donations from Ontario and B.C. districts to Albertan candidates, over all three election periods. Taken together, our results suggest that out-of-district political giving in Canada is not a singular phenomenon. Donations to EDAs are concentrated among a relatively small number of affluent source districts and follow broad population and wealth gradients, whereas candidate-directed donations exhibit stronger signatures of ethnic, spatial, and partisan motivations.

Our findings are accompanied by some notable limitations. First, a non-negligible proportion (26%) of the total contributed amount addressed to candidates and district associations during our study period was either location-anonymized or malformatted, meaning that these could not be assessed as out-of-district. However, a comparison of located and unlocated contributions see Appendix D revealed that these missing donor sums had similar distributions to the study data across factors such as recipient party, district, and entity, suggesting that their exclusion introduced little systematic bias into our study dataset. Previous North American studies using similar datasets chose to omit contributions under $200 entirely (Garnett 2024; Gimpel et al. 2008). By retaining these smaller donations in our study and demonstrating that unlocated records are similarly distributed across available factors, our approach offers a more complete picture of the Canadian local funding landscape.

Second, our focus on analyzing out-of-district contributions meant excluding donations addressed to national entities (parties and leadership contestants). As shown in Section 3.2, contributions of this type represented a large majority (86%) of total contributions during our study period. Although necessary given our study topic, these exclusions limit the generalizability of our study; the average Canadian donor cannot actually be assessed as an out-of-district or in-district giver.

Third, our chosen study period (October 2015 to April 2024) awkwardly truncates the available data in the middle of the period between the 44th and 45th general elections. This was done in an attempt to restrict our analysis to a single, static set of electoral districts (FEDs). While we succeeded in excluding non-2013-order FEDs, we cannot guarantee that all valid 2013-order FED entries were captured[8], since the adoption of new FEDs is a multi-year project and timelines can differ drastically between locations (Elections Canada 2025b). Further, our chosen date range results in the third non-election period of our analysis being incomplete,

[8]Over all entries in the dataset which had recipients belonging to 2013-order districts (2004 - present), only 0.5% were not captured by our chosen range.

so we cannot claim to have studied three full parliaments. However, we did not identify any major differences in donation fluxes between interelection periods, and mainly found differences between election and non-elction periods, so the impact of this loss is likely minor.

Appendix

A Background on Relevant Canadian Electoral Regulations

A.1 Individual Contribution Limits

The *Canadian Elections Act* imposes individual contribution limits on donors seeking to contribute to one of the following registered, federal political entities: political parties, electoral district associations (EDAs), candidates, party nomination contestants, and party leadership contestants (Elections Canada 2026a). These limits are shown in Table 11 and Table 12[9]. A full definition for each of these is available at Elections Canada (2026d). Regulated expenses for any of the entities themselves cannot exceed specific values, which are determined on an annual basis, and are different for each riding (see Elections Canada 2026b). Leadership contestants are a notable exception, with no campaign spending limits (Elections Canada 2026b). Individual contributions to registered third-party organizations are not regulated by the *Act*, and were not included in this study (Elections Canada 2026f).

Only Canadian citizens or permanent residents can make contributions (Elections Canada 2026f). Any donor who contributed \$20 or more to any of the five federal political entities listed above must have their name recorded (but not reported to Elections Canada). Donors with contributions exceeding \$200 must also have their names and addresses recorded and reported to Elections Canada. Cash donations above \$20 are prohibited (Elections Canada 2026f).

[9]Contribution limits presented here are only for 2026. They increase by \$25 every year on January 1st.

Table 12: 2026 Limits on Individual Contributions, Loans, and Loan Guarantees, for Politicians Contributing to Their Own Campaigns.

Politician	Own-Campaign Contribution Limit (CAD)
Nomination Contestant	\$2,775
Candidate	\$5,000
Leadership Contestant	\$25,000

Table 11: 2026 Annual Limits on Individual Contributions, Loans, and Loan Guarantees (adapted from Elections Canada website).

Political Entity	2026 Annual Limit (CAD)
To each registered national party	\$1,775
In total, to all EDAs, nomination contestants, and candidates (for each registered party)	\$1,775
In total, to each independent or non-affiliated candidate	\$1,775

A.2 Financial Reporting Requirements of Federal Political Entities

All of the information presented in this section can be found on the "Political Financing" page of the Elections Canada website (Elections Canada 2026e). Financial returns represent an account of an entity's inflow (i.e. contributions, loans or transfers) and outflow (expenses). At minimum, all entities must submit financial returns annually (for parties and EDAs), or within 4 to 8 months after an electoral event (for parties, candidates, third parties, leadership contestants, and nomination contestants exceeding \$1000 in contributions or expenses). Returns with large transaction amounts must be accompanied by an external audit. Entities such as parties and EDAs may be required to file reports on a quarterly basis. Some (but not all) returns are reviewed by Elections Canada auditors. Risk-based auditing is used to determine which files will be audited and what level of audit will be performed (Elections Canada 2026c).

A.3 Electoral Redistricting

Electoral district boundaries are subject to change every 10 years, as required by the *Constitution of Canada* (Elections Canada 2025b). The 2013 representational order consisted of 338 districts, and the 2023 representational order consisted of 343 districts. The boundaries of existing districts are subject to change as well. The 2013 representational order was decidedly in effect between the calling of the 42nd general election on October 2, 2015, and the first dissolution of Parliament occurring after April 22, 2024 (Elections Canada 2023).

B Donations Data Pipeline

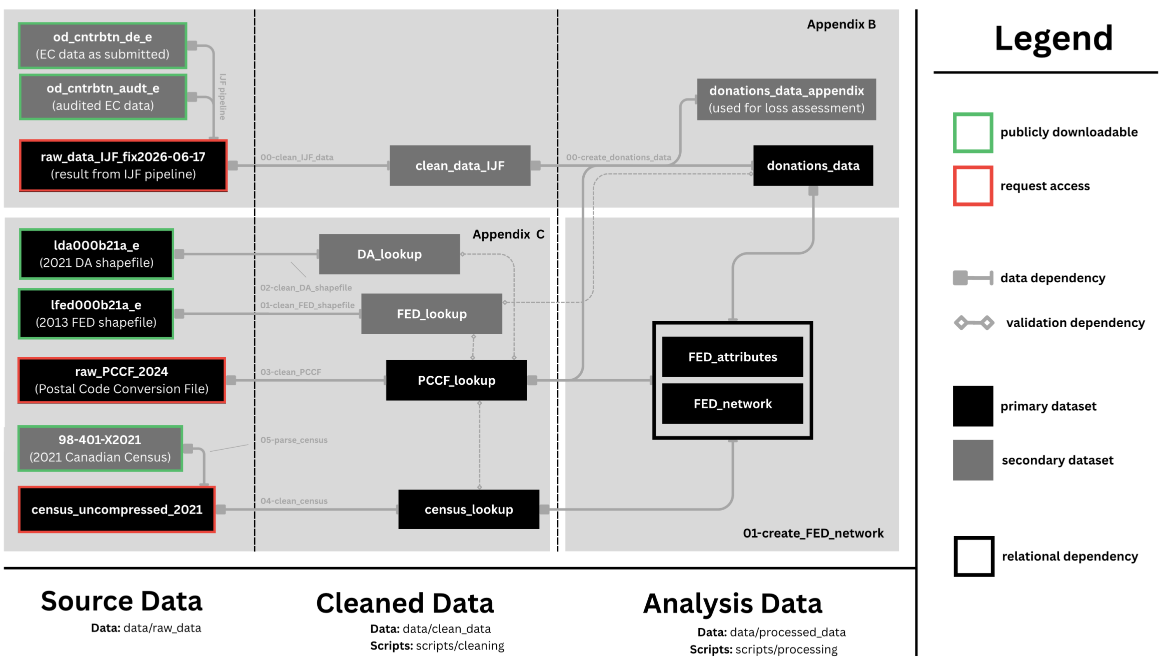

Figure 10: Flow diagram illustrating the full data pipeline used throughout this analysis. All scripts referenced in this figure can be accessed at https://github.com/bennypk1/geolocating-political-donations. Diagram made using the Canva graphic design software.

In this section we provide a plain-language walkthrough of the three major cleaning and validation scripts used to transform contributions data from Elections Canada into the dataset used throughout our analysis. These scripts rely on other datasets, cleaned separately, which are documented in Appendix C. The full pipeline, including the steps described in Appendix C, is illustrated in Figure 10.

B.1 Investigative Journalism Foundation Pre-Cleaning

The original contribution datasets used in this study can be found on Elections Canada's "Open data on contributions" page, under the heading "Data sets updated" (Elections Canada 2025a). Two products are offered. The first is a dataset of contribution reports filed by all Canadian registered federal political entities see list of these in Appendix A.1 from January 2004 to present (Elections Canada 2025a). The second is an audited version of the first. Due to pre-screening policies and administrative lag, only some of the first dataset is audited. In our case, the audited version of the dataset has only 61.7% as many rows as the full un-audited data (n = 10095198 ; both accessed on May 5, 2026). Both datasets are updated on a weekly basis by Elections Canada (Elections Canada 2025a), as new audits and financial reports are filed.

Staff at the Investigative Journalism Foundation created a data pipeline to aggregate, de-duplicate, and clean these datasets. Both audited and un-audited versions of the same record were available for most records. To avoid duplicating these, audited and un-audited versions of the same record were linked by assuming that rows sharing the same donation date, electoral event, recipient, and recipient political party referred to the same entry. We note that this linkage approach carries the small risk that two distinct donations to the same recipient on the same date are incorrectly treated as duplicates. The pipeline preferentially keeps audited records when they are available.

Further de-duplication was effectuated differently based on the type of contribution being recorded. For loans, sums reported during an Annual filing might appear in a Quarterly report filed in the same year. In such cases, Annual records were preserved and Quarterly records were discarded. For all other types of contributions, only true duplicates (i.e. rows with identical entries identical in every field) were discarded. Additionally, some entries represented aggregated sums of contributions from anonymous donors. For these, Elections Canada reports a cumulative total per recipient Annually and Quarterly, meaning that the same aggregate sum could appear twice in the same year. To avoid double-counting these, only the largest reported aggregate sum per recipient per year was retained. Finally, a few smaller cleaning procedures were applied to the columns themselves; Table 13 shows where each of the columns present in the IJF data were drawn from columns at the source, and any notable procedures applied to them. Together, these changes reduced the dataset from 10095198 to 7479577 rows (as of June 17, 2026).

B.2 Initial Cleaning of IJF Data

We applied further cleaning and filtering steps to the data supplied by the Investigative Journalism Foundation. First, we removed redundant columns (e.g. data had columns `year` and `donation_year` which were identical for every entry) and columns used for internal consistency (e.g. the date the entry was added by the organization). Next, we removed entries with malformatted dates, or disagreements between the year and date columns. Less than 0.001% of entries were discarded at this step, all corresponding to donations dated in later months of 2026. We then filtered for only donations stated to have been made by individuals. Less than 0.4% of the data was removed at this step, the overwhelming majority of which were records dated before 2006, outside the scope of our study. Next, we filtered for donations where the sum of monetary and non-monetary contributions was present and positive. Less than 0.02% of the data was removed at this step. We then removed entries where the donor or recipient name was either missing or shorter than 4 characters (after trimming whitespaces). Less than 0.001% of data was removed at this step, most of which also had missing donor locations. Finally, we removed individual donations with amounts larger than $25000. Here we were careful to exclude entries that represented aggregated sums of anonymous contributions, which were flagged by searching for entries where `donor_full_name` started with "Contribut". Less than 0.001% of the data was removed at this step. In total, 0.4% of entries (n ~ 30,000) were discarded in this section.

Donor postal codes were extracted from the `donor_location` column. Canadian postal codes always follow the format A0A0A0, where 0 is a digit and A is a letter (excluding I, O, D, F, Q, or U). Also, they may not start with W or Z (Canada Post 2026). These facts were used to verify the structure of extracted postal codes, and apply the following imputations:

- "O"s in a number slot were taken to mean "0"
- "I"s in a number slot were taken to mean "1"
- "!" in a number slot were taken to mean "1"

After imputations we were able to recover 99.89% of individual donor postal codes. The remaining 0.11% were kept, with their postal codes set to `NA`. We note that while these postal

Table 13: Description of where the columns in the IJF data were derived from the source, and any cleaning procedures applied. Variables added for internal database consistency (e.g. processing dates) and redundant variables (e.g. donation year) are not shown here.

IJF Column	EC Source Column(s)	Notes
donor_full_name	Contributor name ; Contributor last name ; Contributor first name ; Contributor middle initial	Concatenated source columns, delimited by commas.
donor_type	Contributor type	No changes from source.
donor_location	Contributor City ; Contributor Province ; Contributor Postal code	Concatenated source columns, delimited by commas.
donation_date	Contribution Received date ; Fiscal/Election date	Used 'Contribution Received date' when available, filled with Fiscal/Election date otherwise. Removed invalid dates.
electoral_event	Electoral event	Leadership and Nomination contestant were blank, nothing else was. Unclear why. Otherwise, no changes from source.
amount_monetary	Monetary amount	No changes from source.
amount_non_monetary	Non-Monetary amount	No changes from source.
recipient	Recipient ; Recipient last name ; Recipient first name ; Recipient middle initial	Concatenated source columns, delimited by commas.
electoral_district	Electoral District	No changes from source.
political_entity	Political Entity	No changes from source.
political_party	Political Party of Recipient	No changes from source.

codes are guaranteed to be structurally correct, we did not validate whether the codes matched their associated provinces, cities, or were actually used. The effective validity of these postal codes is verified in Appendix B.3. 0.35% of entries (n ~ 26,000) represented aggregated sums of anonymous contributions, and therefore did not have associated postal codes. Despite their scarcity, these amounted to 24.1% of the total contributed amount in the dataset. These were kept at this stage.

We also performed simple string cleaning procedures such as whitespace stripping and entity matching (e.g. United Party of Canada (UP) and United Party of Canada clearly refer to the same political party). Notably all donation recipients in the datset had associated electoral districts (apart from leadership contestants and registered parties, since these are non-local entities) and political parties. Spot checking revealed no district or party attribution errors among a small set of recipients, however we could not completely confirm that recipient districts or parties were correct.

B.3 Clean IJF to `donations_data`

In this section we describe how our primary analytical dataset (`donations_data` in the GitHub repository) was derived from the cleaned IJF data, and augmented (to include the donor's district) using a custom postal code conversion file (PCCF) described in Appendix C.1.

First, we restricted the dataset to only those entries with donation dates between October 4, 2015, and April 22, 2024, capturing the effective period of the 2013 representational order see Appendix A.1. This period captures 61.2% of the entire contributions dataset. Next, we removed entries where the recipient's district was not one of the 338 possible districts in the 2013 representational order, referencing our modified PCCF (0.002% of cases). Next, we removed entries where the reported electoral event did not happen within our date range (e.g. 39th general election was in 2006). Although it is possible that these were retroactively added for past events, less than 0.001% of the data had this issue, so their removal did not impact our analysis. Next, we removed donations addressed to leadership contestants for parties where no such contest occurred during our study period (0.0001% of cases). Nomination contests had relatively tiny contribution amounts (0.08% by count and 0.3% of total contributed amount at this stage), and were therefore not validated in this way. 3% of donations (7% of summed total) received during general election periods had receipt dates outside the stated period (usually dated after the writ period had ended, not before it started). These were not removed due to their non-trivial abundance, however many analyses and visualizations split the data into donations received during vs. outside writ periods. In those cases, these data are considered to have been received outside a writ period. All such anomalies originated from financial reports filed by electoral candidates.

Donor postal codes were matched to their corresponding census dissemination areas (DAs) and federal electoral districts (FEDs) using our modified PCCF file. Using this file, 1.11% of donor postal codes could not be identified, representing 1.33% of the total amount contributed by entries with structurally valid postal codes. Since matching donors to districts is necessary to assess whether a donation is out-of-district, these unmatched codes, along with entries missing postal codes entirely were removed from the study dataset. Overall, 22.5% of the total contributed amount could not be donor-geolocated (26% for donations addressed to local entities). Figure 11 illustrates what proportion of the total amount donated during our study period was successfully geolocated, broken down by cause of missingness. The potential impacts of this unlocated data on the integrity of our study dataset are discussed in Appendix D. In total, 38.78% of data was discarded by date filtering, 0.00237% of entries were discarded for general cleaning, 0.1671% of entries were discarded for missing donor location entirely, and 1.04% of entries were discarded for having invalid postal codes (0.09% due to malformation, and 0.95% due to invalidity).

Finally, columns were renamed, merged, and variables were joined to improve the dataset's usability. The resulting schema is shown in part in Section 3.1. We provide a list of notable operations here below:

- Monetary (99.8%) and non-monetary (0.2%) contributions were joined, and analysis was carried out on the total.
- Specific electoral events were grouped into the categories: Leadership contest, Nomination contest, General election, By-election, and Non-electoral (i.e. annual or quarterly filings).
- We added a variable to indicate during which writ period (if any) a donation was received, based on `donation_date`. Writ periods were taken to be August 2 to October 19 (2015), September 11 to October 21 (2019) and August 15 to September 20 (2021). By-elections are excluded from election vs. non-election analyses, however these account for less than 0.1% of the total contributed amount during the study period (Table 14).

Table 14: Breakdown of contributions by electoral event (as indicated on the financial report) during the study period. This table only includes donations whose donors were successfully geolocated.

Source	Amount	Count	Proportion (of Amount)
Non-electoral	\$557,830,730	4,162,516	88.8%
Leadership contest	\$47,423,142	306,326	7.5%
Nomination contest	\$2,378,332	3,299	0.4%
General election	\$20,051,698	32,473	3.2%
By-election	\$673,287	1,128	0.1%
Total	\$628,357,188	4,505,742	100.0%

Table 15: Breakdown of contributions by whether or not they were received during the writ period of a general election, during the study period. This table only includes donations whose donors were successfully geolocated. In the main text, we call this separation "on-cycle" vs. "off-cycle" donations.

Source	Amount	Count	Proportion (of amount)
During general election writ period (on-cycle)	\$87,053,110	390,664	13.9%
NA	\$541,304,078	4,115,078	86.1%
Total	\$628,357,188	4,505,742	100.0%

Table 16: Breakdown of contributions by whether the recipient is a national-level (registered parties, leadership contestants) or local (candidates, EDAs, nomination contestants) political entity, during the study period. This table only includes donations whose donors were successfully geolocated.

Source	Amount	Count	Proportion (of amount)
National-level entity	\$532,130,503	4,066,335	84.7%
Local entity	\$96,226,685	439,407	15.3%
Total	\$628,357,188	4,505,742	100.0%

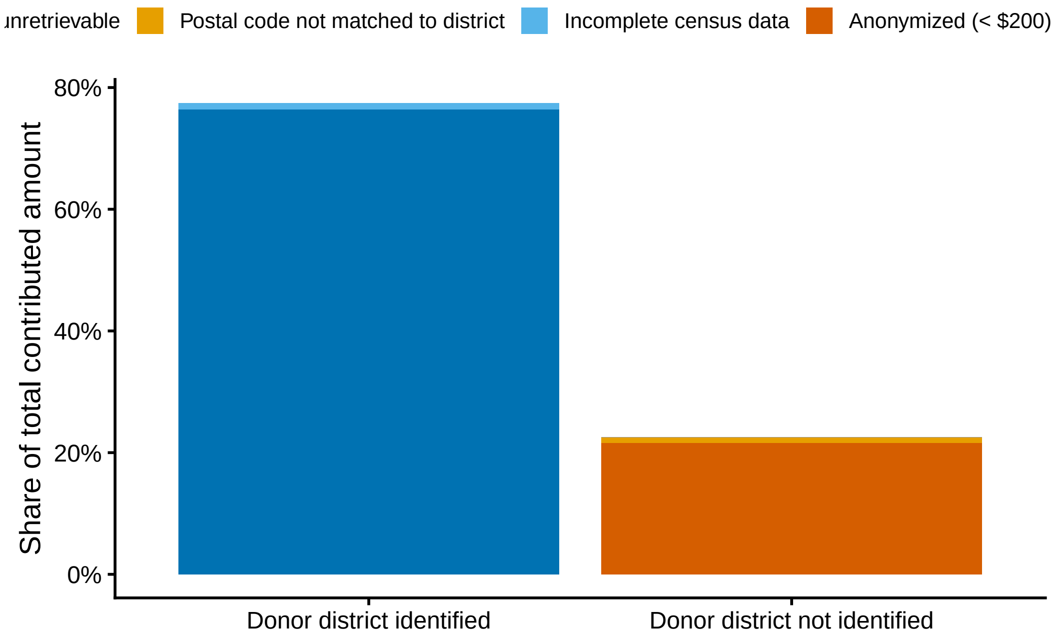

Figure 11: Geolocation success for all contributions to federal political entities received from 2015 to 2024.

C Creating Supplementary Datasets

In this section we describe how we created two custom lookup tables that the cleaning scripts in Appendix B rely on; one to convert between different levels of spatial hierarchy, and a second to retrieve specific socio-demographic statistics at the census dissemination area. Their place in the full pipeline is illustrated in Figure 10.

C.1 Spatial Lookup Table (`PCCF_lookup` on GitHub)

The purpose of the table constructed in this section is to provide unambiguous links between any administrative area of interest and another at a larger scale (e.g. postal code to district, dissemination area to province, etc.). This custom table is a version of Statistics Canada's 2024 postal code conversion files (PCCF), modified to resolve many-to-many spatial overlaps between postal codes, dissemination areas (DAs), and electoral districts (FEDs). The PCCF we used provides linkages between all registered Canadian postal codes (accurate as of June 2024) and various administrative divisions, including the dissemination areas (DAs) of the 2021 Census, and the federal electoral districts (FEDs) of the 2013 representational order (Statistics

Canada, Geography Division 2025). These sets of DAs and FEDs are the same versions we chose for our study. Access to this PCCF was granted to us by the University of Toronto.

To resolve cases where one postal code straddled multiple dissemination areas, we used a single link indicator (SLI) to establish a one-to-one relationship between postal codes and DAs, available in PCCF files by default (Statistics Canada, Geography Division 2025). Statistics Canada determines SLIs by estimating the population that resides in each DA within the postal code area (based on 2021 Census population counts) and assigns the code to the one with a higher population (Statistics Canada, Geography Division 2025). This procedure removed 35% of links present in the original PCCF, however no single postal code was removed from the file entirely. A small number of postal codes (0.7%) were removed for missing either an associated DA or FED. All postal codes had associated provinces. In addition, 1.6% of dissemination areas in the dataset straddled multiple district boundaries. To resolve this, for each overlap case, we (1) counted how many postal codes within the dissemination area linked to each candidate FED, and (2) assigned the entire DA, and all of its postal codes to the FED holding the plurality of those postal codes, discarding the postal codes that mapped to any minority FED. Ties between FEDs were broken deterministically by FED UID. This resolution step resulted in the loss of roughly 0.5% of postal codes. Applied to the contributions dataset used in this study, these resolutions increased the share of entries whose donor postal codes could not be matched to the PCCF from 0.46% (n = 20,800) to 0.95% (n = 43,059).

As a final step, we ran a suite of tests, primarily to confirm that the resolved table formed a strictly nested hierarchy: each postal code mapped to a single DA, each DA or postal code to a single FED, and each FED, DA, or postal code to a single province. We also verified that every postal code appeared exactly once, and that they were correctly formatted following Canada Post's guidelines (Canada Post 2026). We also confirmed that all DA and FED UIDs could be cross-referenced with Statistics Canada's 2021 DA and 2013 FED shapefiles (Statistics Canada 2022a), and that all 338 districts and 13 provinces remained represented. Because of these tests, this spatial lookup table was treated as an authoritative list of valid postal codes, dissemination areas, districts, and provinces, which greatly simplified our analysis.

C.2 Census Lookup Table (`census_lookup` on GitHub)

The purpose of the table constructed in this section is to provide specific socio-demographic variables (listed in Table 17) for every dissemination area (DA) present in the spatial lookup table (described in Section C.1). Socio-demographic information at this spatial scale was sourced from the 2021 Census, and specifically from the "Canada, provinces, territories, census divisions (CDs), census subdivisions (CSDs) and dissemination areas (DAs)" available for download online (Statistics Canada 2022b). We chose the 2021 Census over the 2016 Census since the midpoint of our time period (2019) is closer to 2021 than 2016. We chose the dissemination area, an administrative region that houses roughly 400 to 700 people, since it is the smallest spatial scale where each of our desired statistics are recorded (Statistics Canada 2021). All DAs listed in our spatial lookup file see Appendix C.1 were present in the Census dataset we used. Since these DAs are the only ones we would ever use for analysis, the other 15.1% of DAs were dropped.

3.2% of remaining DAs were missing at least one census variable: 0.05% were missing all demographic variables, 0.43% only had population, area, and density, 0.12% were only missing the 25% sampled variables (i.e. ethnicity and education level), and 2.62% had all variables recorded except household income. Notably, all entries' missingness could be grouped into five strictly hierarchical stages, summarized in Table 18. Entries with missing variables were not dropped, but were instead flagged so that they could be easily identified during analysis. We also converted ethnicity and education counts to proportions.

D Characterizing Unlocatable Contribution Data

22.5% of the total contributed amount (and 26% of total contributions to local political entities) during our study period could not be donor-geolocated. For 95.8% (93.4% for local entities) of this amount, donor locations were intentionally left anonymous. The remaining entries (0.94% overall and 1.7% for local entities, across the entire dataset) could not be donor-geolocated due to malformatted locations or unmatched postal codes. This includes cases where postal codes were sacrificed to resolve the many-to-many relationship between dissemination areas and electoral districts see Appendix C.1. We characterize intentionally and unintentionally missing data separately. We restrict this section to donations addressed to candidates and district associations, since these are the only ones that could have been analyzed as out-of-district.

Table 17: Census IDs, descriptions, and sample sizes for each census variable used in this study. Variables with a 25% sample size were recorded for only 25% of the population, to preserve privacy.

Characteristic ID	Description	Sample Size
1	Population, 2021	100%
6	Population density (per square kilometres)	100%
7	Land area (square kilometres)	100%
39	Average (mean) age	100%
243	Median household income (total, in 2020)	100%
1683	Total population in private households	25%
1684	Total visible minority population	25%
1685	Total South Asian population	25%
1686	Total Chinese population	25%
1687	Total Black population	25%
1688	Total Filipino population	25%
1689	Total Arab population	25%
1690	Total Latin American population	25%
1691	Total Southeast Asian population	25%
1692	Total West Asian population	25%
1693	Total Korean population	25%
1694	Total Japanese population	25%
1998	Total population aged 15+	25%
2001	Total population with a postsecondary certificate	25%

Table 18: Structure of missing census data.

Missing Stage	Count	% of Total
complete	47593	96.78%
missing_income_only	1288	2.62%
missing_25pct	57	0.12%
mostly_missing	211	0.43%
missing_population	26	0.05%

D.1 Anonymous Contribution Sums

24.28% of the total contributed sum addressed to candidates and district associations was presented as aggregated sums of many anonymous contributors. As such, neither a specific location, nor a specific date could be identified. However, every aggregated sum always had an associated electoral event, recipient type (candidate or district association), recipient party, and recipient district. In Figure 12, we compare the marginal distributions of aggregated contributions across these factors with the same distributions in the local contributions data used in our study (recipient district was grouped by province to allow for effective visualization). We observe that aggregated sums are similarly distributed to donor-geolocated data, across all factors. In each case, the rank of a category is the same for the aggregated and study data, and relative proportions deviate at most by 10%. This provides some evidence that lacking these contributions does not substantially bias our estimates of out-of-district donation rates.

D.2 Unsuccessfully Located Individual Contributions Sums

1.7% of the total amount contributed to candidates and district associations during our study period represented individual contributions where we were unable to identify the donor's district. This was either due to (1) location was not recorded at source, (2) postal code was malformatted at source, or (3) postal code could not be matched to our spatial lookup table see Appendix C.1. Figure 13 compares the marginal distributions of aggregated contributions across five factors with the same distributions in the local contributions data used in our study (recipient district was grouped by province to allow for effective visualization). This missing data exhibits larger deviations than the aggregated sums shown in Appendix D.1. However, this data is still similarly distributed to donor-geolocated data overall, with relative proportions

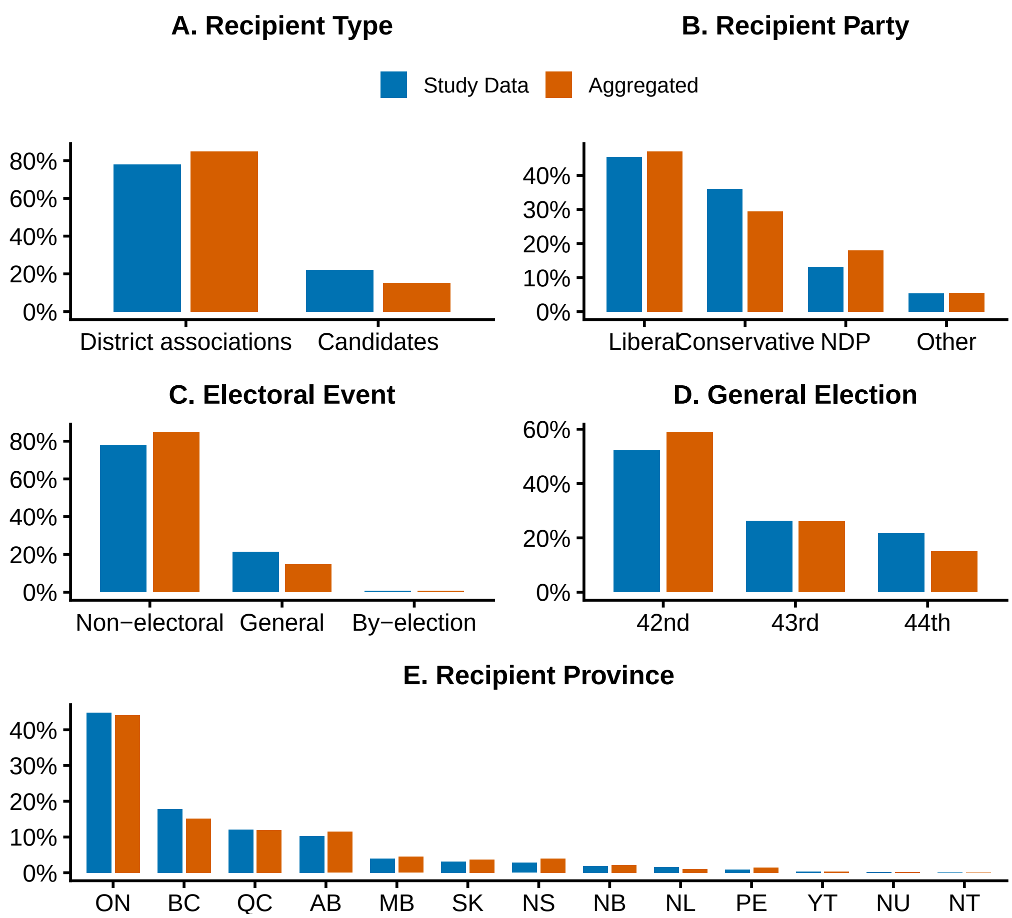

Figure 12: Comparing the marginal distributions of successfully geolocated donations (blue) vs. anonymized contribution sums (red) across multiple factors (labelled in bold in the subplots themselves). Only donations to candidates and district associations (EDAs) are considered. The breakdown of the total amount contributed across each factor is plotted on the y-axis. In each sub-plot, blue bars heights sum to 100%, and red bars heights separately sum to 100%.

deviating by at most 12%. Further, these data only represent 1.7% as opposed to the 24% in the case of aggregated sums, so biases incurred by dropping them are less impactful to the study.

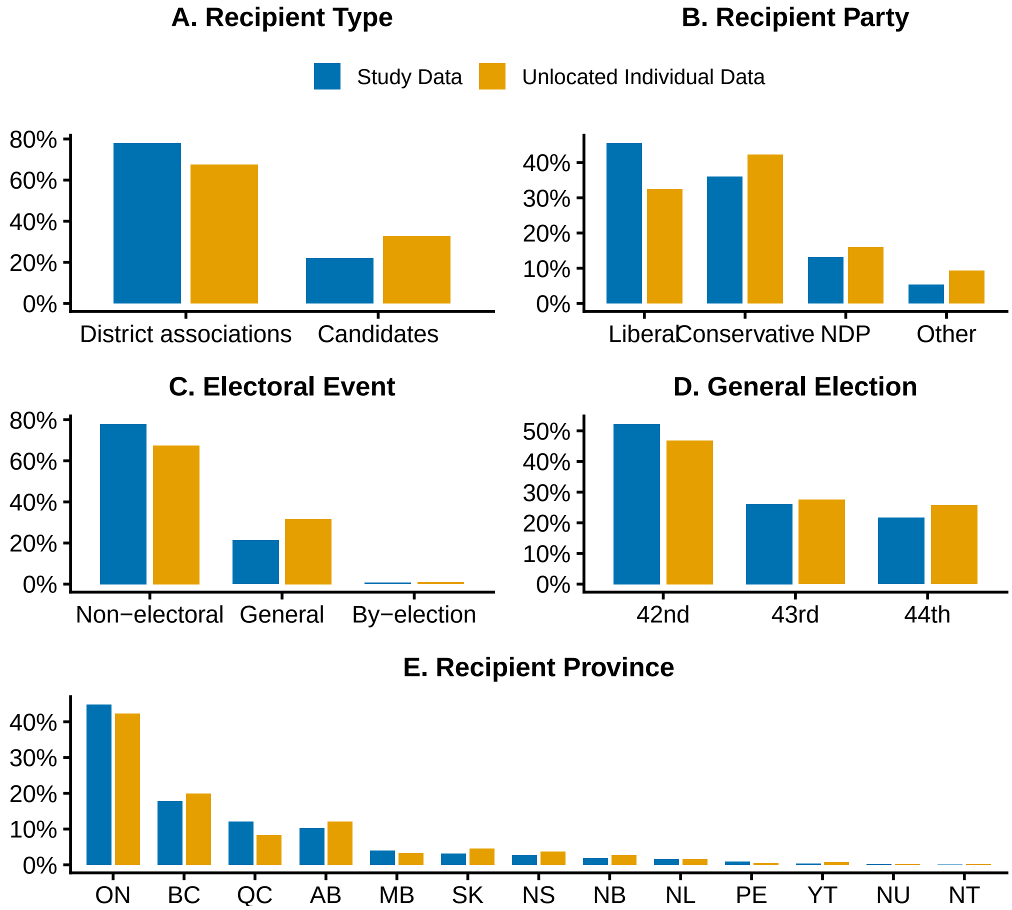

Figure 13: Comparing the marginal distributions of successfully geolocated donations (blue) vs. individual contributions that could not be geolocated (red) across multiple factors (labelled in bold in the subplots themselves). Only donations to candidates and district associations (EDAs) are considered. The breakdown of the total amount contributed across each factor is plotted on the y-axis. In each sub-plot, blue bars heights sum to 100%, and red bars heights separately sum to 100%.

E Modifications to Data for Figures & Alternate Visualizations

- **Longitudinal Visualizations:** All types of donations occurred at much higher rates in December compared to other months, regardless of year, recipient type, party affiliation, etc. There are two likely causes for this; a clerical error at the source, or, the IJF frequently imputes the fiscal/election date in the absence of donation receipt date. This did not impact our analysis, however December donations in all Figures presented below were imputed using their neighboring November and January values, and their recorded monetary amounts were redistributed across their associated years so that nothing was lost. This served to reduce visual clutter from large, phantom peaks at the end of each year.

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